Applied Economic Forecasting

7. Vector Autoregressive Models (Part I)

1. Motivation
2. The VAR Model
3. Stationarity and the VAR
4. Lag Selection
5. Estimation
6. Granger Causality
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Section 1

Motivation

Why Single-Equation Models Fall Short

So far, we have built forecasting models **one equation at a time.**

Consider a policy question: *How does a surprise increase in the federal funds rate affect output and inflation?*

The single-equation approach runs into trouble immediately:

- A model for GDP growth may include the interest rate as a regressor — but the interest rate itself responds to GDP growth.
- A model for inflation may include GDP growth — but GDP growth responds to inflation.
- Each variable is **endogenous** to the system.

The simultaneity problem

When variables are jointly determined, treating any one of them as strictly exogenous leads to omitted variable bias and invalid inference.

A Systems Approach

The natural fix: Model all variables jointly.

Rather than

$$\text{GDP growth}_t = f(\text{inflation}_t,\ \text{interest rate}_t, \ldots)$$

we want a framework that lets each variable depend on **lags of all variables in the system** — with no variable treated as exogenous by assumption.

This is precisely what a **Vector Autoregression (VAR)** does.

A Systems Approach

VAR models were introduced by **Christopher Sims (1980)** as a reaction to the large, heavily restricted simultaneous equation models dominant at the time.

Sims's critique

Exclusion restrictions in traditional macro models were "incredible" — imposed for identification convenience, not from theory. Let the data speak instead.

What We Can Do with a VAR

A well-specified VAR is a workhorse for:

- **Forecasting** — multi-step forecasts for a system of variables that account for dynamic interactions
- **Granger causality** — testing whether past values of one variable help predict future values of another
- **Impulse Response Functions (IRFs)** — tracing the dynamic effect of a shock in one variable on all others
- **Forecast Error Variance Decomposition (FEVD)** — measuring how much of the forecast uncertainty in one variable is attributable to shocks in each variable of the system

Section 2

The VAR Model

The VAR(p) Model

Definition

A VAR of order p, written VAR(p), is a system of K equations in which each variable is regressed on p of its own lags and p lags of every other variable in the system.

For a system of K variables collected in the vector $\mathbf{y}_t = (y_{1t}, y_{2t}, \ldots, y_{Kt})'$:

$$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1\mathbf{y}_{t-1} + \mathbf{A}_2\mathbf{y}_{t-2} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \boldsymbol{\varepsilon}_t$$

where

- $\mathbf{c}$ is a $K \times 1$ vector of intercepts
- $\mathbf{A}_i$ is a $K \times K$ matrix of coefficients at lag i
- $\boldsymbol{\varepsilon}_t \sim (0, \boldsymbol{\Sigma})$ is a vector of white noise errors with covariance matrix $\boldsymbol{\Sigma}$

A Bivariate VAR(1): Written Out

Let $\mathbf{y}_t = (y_{1t}, y_{2t})'$. A VAR(1) is:

$$y_{1t} = c_1 + a_{11} y_{1,t-1} + a_{12} y_{2,t-1} + \varepsilon_{1t}$$
$$y_{2t} = c_2 + a_{21} y_{1,t-1} + a_{22} y_{2,t-1} + \varepsilon_{2t}$$

- a_{12}: the effect of a lagged change in y_2 on y_1
- a_{21}: the effect of a lagged change in y_1 on y_2
- ε_{1t} and ε_{2t} may be **contemporaneously correlated**: $\mathrm{Cov}(\varepsilon_{1t}, \varepsilon_{2t}) = \sigma_{12} \neq 0$

The reduced form

This system is called the **reduced form VAR** — each equation contains only lagged (predetermined) variables on the right-hand side. No endogeneity problem: OLS equation-by-equation is consistent and efficient.

Our Running Example

A three-variable macro system, 1960Q2–2023Q4

We work with the classic Sims-style system:

$$\mathbf{y}_t = \begin{pmatrix} g_t \\ \pi_t \\ r_t \end{pmatrix}$$

where

- g_t — real (annualized) GDP growth: $400 \times \Delta \ln(\text{RGDP}_t)$,
- π_t — CPI inflation (annualized): $400 \times \Delta \ln(\text{CPI}_t)$,
- r_t — federal funds rate (%)

Note on ordering

The ordering of variables matters once we move to structural identification. We return to this in Part 2.

Section 3

Stationarity and the VAR

Stationarity in a VAR System

The stationarity requirement we applied to univariate models extends naturally to the multivariate setting.

Stability condition

A VAR(p) is **stable** (stationary) if all roots of

$$\det\Big(\mathbf{I}_K - \mathbf{A}_1 z - \mathbf{A}_2 z^2 - \cdots - \mathbf{A}_p z^p\Big) = 0$$

lie **outside** the unit circle.

In practice:

- We test each series for unit roots individually before estimation.
- If series are integrated, we either difference or test for cointegration.
- If cointegration is present, a VECM is preferred over a VAR in differences.

Levels vs. Differences: A Practical Note

The textbook workflow says: test, difference if needed, then estimate.

The practitioner's view (Sims, Stock, Watson) is more nuanced:

- Estimating a VAR in **levels** with integrated variables yields consistent coefficient estimates and valid forecasts — even without differencing.
- Unit root pre-tests have **low power** and can distort downstream inference.
- Differencing throws away long-run information and is a **misspecification** when cointegration is present.

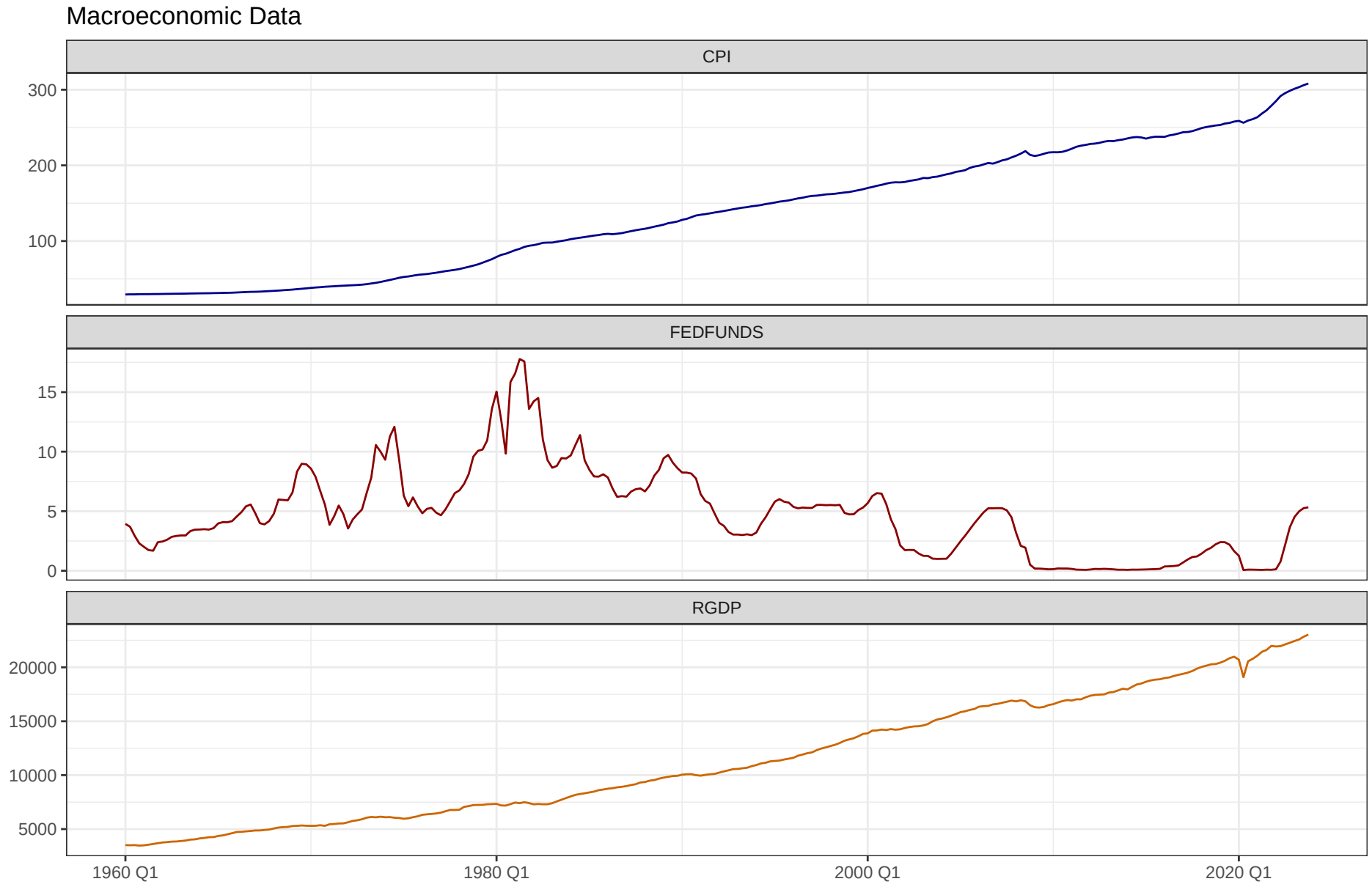
Our approach

We will follow the textbook workflow in this module. Test each series, transform as needed, estimate the VAR, then ask whether a long-run relationship may have been discarded — which motivates the VECM.

Step 1: Visualize the Series

```r
macro.raw <- read.csv("macro_data.csv") %>%
  mutate(Quarter = yearquarter(Quarter)) %>%
  tsibble(index = Quarter)

macro.raw %>%
  pivot_longer(-Quarter, names_to = "Series") %>%
  ggplot(aes(x = Quarter, y = value, col = Series)) +
  geom_line() +
  facet_wrap(Series ~ ., scales = "free_y", dir = "v") +
  labs(title = "Macroeconomic Data", x = NULL, y = NULL) +
  theme(legend.position = "none") +
  scale_color_manual(values = c("blue4", "darkred",
                                "darkorange3"))
```

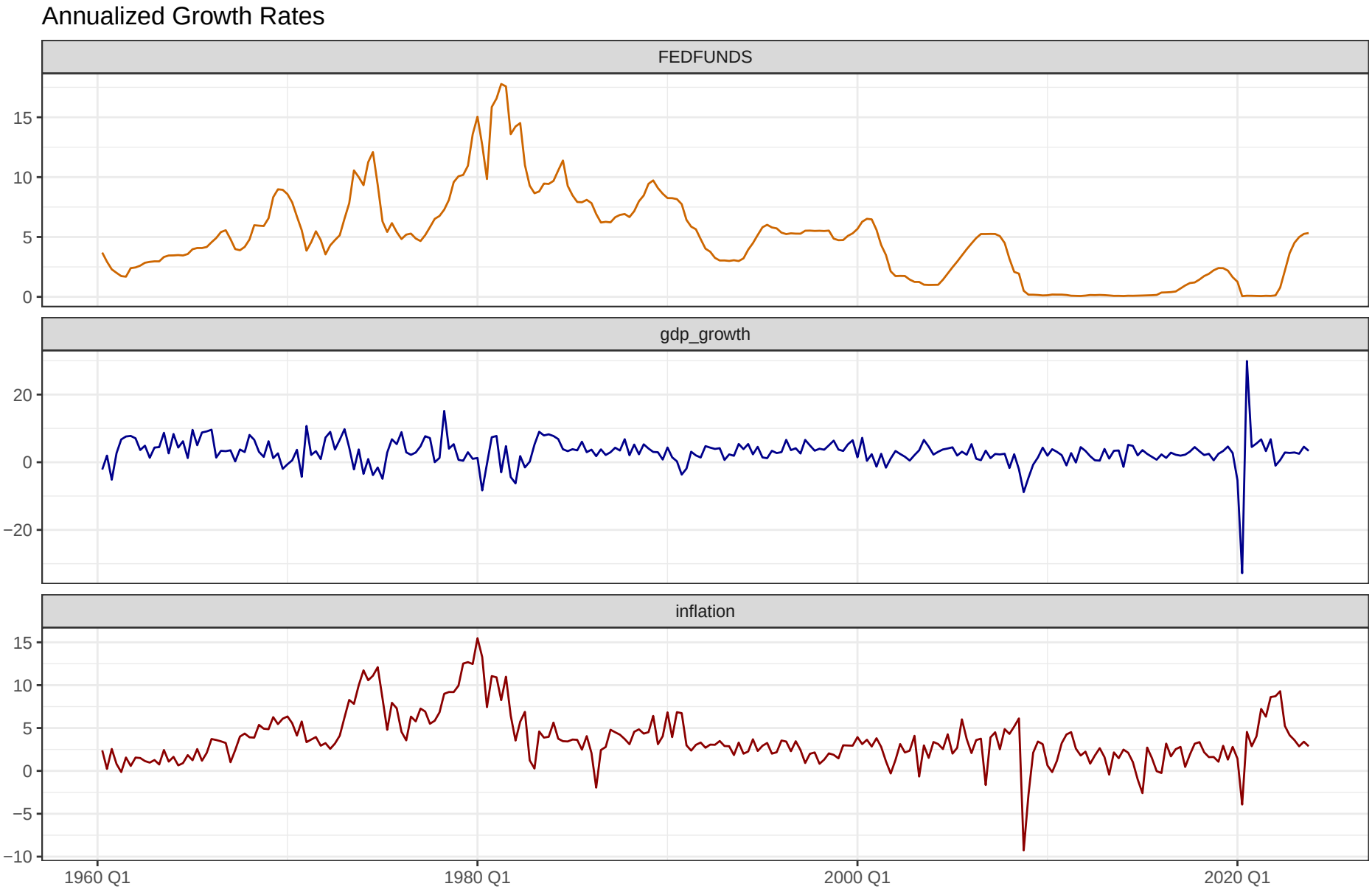
Step 1: Visualize the Series



Annualized Growth Rates

```
macro <- macro.raw %>%
  mutate(gdp_growth = 400 * difference(log(RGDP)),
         inflation  = 400 * difference(log(CPI))) %>%
  slice(-1)

macro %>%
  pivot_longer(c(gdp_growth, inflation, FEDFUNDS), names_to = "Series") %>%
  ggplot(aes(x = Quarter, y = value, col = Series)) +
  geom_line() +
  facet_wrap(Series ~ ., scales = "free_y", dir = "v") +
  labs(title = "Annualized Growth Rates", x = NULL, y = NULL) +
  theme(legend.position = "none") +
scale_color_manual(
  values = c("gdp_growth" = "blue4",
             "inflation" = "darkred",
             "FEDFUNDS" = "darkorange3"),
  labels = c("gdp_growth" = "GDP Growth",
             "inflation" = "Inflation",
             "FEDFUNDS" = "Fed Funds Rate")
)
```

Annualized Growth Rates



Step 2: Unit Root Tests

We test each series using the Augmented Dickey-Fuller (ADF) test.

H_0 : Unit root (non-stationary) $\qquad$ H_1 : No unit root (stationary)

Decision rule

Reject H_0 if $|\tau_{\text{stat}}| > |\tau_{\text{CV}}|$ — the test statistic is more negative than the critical value.

- Reject $H_0 \Rightarrow$ conclude the series is **stationary**.
- Fail to reject $H_0 \Rightarrow$ conclude a **unit root is present**; differencing required.

Note

The ADF τ statistic is **negative**. More negative = stronger evidence against H_0. Critical values are also negative — reject when $\tau_{\text{stat}} < \tau_{\text{crit}}$.

Step 2: Unit Root Tests

Three specifications in `urca`:

```
ur.df(y, type = "none")     # no constant, no trend
ur.df(y, type = "drift")    # constant only
ur.df(y, type = "trend")    # constant and trend
```

- Use *"drift"* for series with a non-zero mean (GDP growth, inflation).
- Use *"trend"* for series that may trend over the sample (fed funds rate).

Lag selection

Include lags to purge serial correlation from the ADF residuals. A common choice is $\lfloor (T/100)^{1/4} \rfloor \times 4$ lags, or set `selectlags = "AIC"` inside `ur.df()`.

Step 2: Unit Root Tests

```r
# require(urca)
## GDP Growth
ur.df(macro$gdp_growth, type = "drift",
      lags = 4) %>% summary()

## Inflation
ur.df(macro$inflation, type = "drift",
      lags = 4) %>% summary()

## Fed Funds Rate
ur.df(macro$FEDFUNDS, type = "trend",
      lags = 4) %>% summary()
```

```
## Critical values for GDP growth:
##  -3.44 6.47 -2.87 4.61 -2.57 3.79
##

## Test statistic for GDP growth:
##  -6.534 21.34
##
##  Test Stat for inflation:
##  -3.137 4.933
##
##  Test Stat for fed funds rate:
##  -3.323 3.721 5.556
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -14.418  -0.783   0.123   1.016   5.803
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.0167     0.1237    0.13   0.8927
## z.lag.1      -1.9837     0.2185   -9.08   <2e-16 ***
## z.diff.lag1   0.6117     0.1893    3.23   0.0014 **
```

Step 2: Unit Root Tests — Results

Summary of ADF tests

Series	**Spec.**	τ_{stat}	**5%**	**10%**	**Conclusion**
GDP Growth	drift	-6.534	-2.87	-2.57	Stationary $I(0)$
Inflation	drift	-3.137	-2.87	-2.57	Stationary $I(0)$ at 5%
Fed Funds	trend	-3.323	-3.42	-3.13	Stationary $I(0)$ at 10%

- Fed funds rate is borderline — we fail to reject H_0 at 5%, but reject at 10%. We treat it as $I(1)$ and difference it.

Proceeding

We first-difference the fed funds rate. We carry this stationary system forward — then ask later whether cointegration means we should use a Vector Error Correction Model (VECM) instead.

Step 2: First-Differencing

```r
macro_st <- ts(cbind(
  gdp_growth  = macro$gdp_growth,
  inflation = macro$inflation,
  d_FEDFUNDS  = diff(macro$FEDFUNDS)),
  start = c(1960, 3), frequency = 4)

## Confirm stationarity on differenced series
ur.df(macro_st[,"d_FEDFUNDS"],
      type = "drift", lags = 4) %>% summary()
```

Section 4

Lag Selection

Step 3: Selecting the Lag Order, p

Information criteria for VARs:

$$\text{AIC} = \ln|\hat{\mathbf{\Sigma}}| + \frac{2}{T} \cdot (K^2 p + K)$$
$$\text{BIC} = \ln|\hat{\mathbf{\Sigma}}| + \frac{\ln T}{T} \cdot (K^2 p + K)$$
$$\text{HQ} = \ln|\hat{\mathbf{\Sigma}}| + \frac{2 \ln \ln T}{T} \cdot (K^2 p + K)$$

where $|\hat{\mathbf{\Sigma}}|$ is the determinant of the residual covariance matrix.

Note

BIC penalizes complexity more heavily and tends toward parsimony. AIC often outperforms BIC for forecasting in small samples. When criteria disagree, BIC is preferred for structural work; AIC for forecasting.

Step 3: Selecting the Lag Order

```r
# require(vars)
vars::VARselect(macro_st, lag.max = 8, type = "const")
```

```
## $selection
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      3      2      1      3
##
## $criteria
##             1      2      3      4      5      6      7
## AIC(n)  3.759  3.702  3.657  3.693  3.690  3.707  3.718  3.
## HQ(n)   3.828  3.822  3.829  3.916  3.964  4.033  4.095  4.
## SC(n)   3.930  4.000  4.083  4.247  4.372  4.517  4.655  4.
## FPE(n) 42.918 40.519 38.751 40.189 40.048 40.761 41.222 42.
```

Step 3: Selecting the Lag Order

How Many Lags Do We Need?

Criterion	**Selected p**	**Tendency**
AIC / FPE	3	Forecast accuracy
HQ	2	Intermediate
BIC (SC)	1	Parsimony

Rather than simply accepting one criterion, we use **diagnostic testing to let the data guide us.**

Strategy

1. Estimate at $p = 1$ (BIC's recommendation — most parsimonious).
2. Run a serial correlation test on the residuals.
3. If residuals are white noise $\Rightarrow$ stop.
4. If serial correlation remains $\Rightarrow$ increase p by 1 and re-test.

This illustrates exactly what we are doing when we "add lags" — include enough dynamic memory to whiten the residuals.

Section 5

Estimation

Step 4: Estimating the VAR

```r
var.fit <- vars::VAR(macro_st, p = 1, type = "const")
summary(var.fit)
# summary(var.fit)$varresult$gdp_growth
```

```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: gdp_growth, inflation, d_FEDFUNDS
## Deterministic variables: const
## Sample size: 254
## Log Likelihood: -1543.815
## Roots of the characteristic polynomial:
## 0.777 0.304 0.104
## Call:
## VAR(y = macro_st, p = 1, type = "const")
##
##
## Estimation results for equation gdp_growth:
## ===========================================
## gdp_growth = gdp_growth.l1 + inflation.l1 + d_FEDFUNDS.l1 + const
##
##               Estimate Std. Error t value Pr(>|t|)
## gdp_growth.l1  -0.0136     0.0609   -0.22   0.8240
## inflation.l1   -0.2689     0.0839   -3.21   0.0015 **
## d_FEDFUNDS.l1   1.3542     0.2971    4.56  8.1e-06 ***
## const           3.9914     0.4340    9.20  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 4.04 on 250 degrees of freedom
## Multiple R-Squared: 0.112,   Adjusted R-squared: 0.101
## F-statistic: 10.5 on 3 and 250 DF,  p-value: 1.53e-06
##
##
## Estimation results for equation inflation:
## ==========================================
## inflation = gdp_growth.l1 + inflation.l1 + d_FEDFUNDS.l1 + const
##
##               Estimate Std. Error t value Pr(>|t|)
## gdp_growth.l1  -0.0539     0.0292   -1.85    0.066 .
## inflation.l1    0.7488     0.0402   18.61  < 2e-16 ***
## d_FEDFUNDS.l1   0.6701     0.1425    4.70  4.2e-06 ***
```

```r
summary(var.fit)$varresult$gdp_growth
```

```
##
## Call:
## lm(formula = y ~ -1 + ., data = datamat)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -34.87  -1.61  -0.08   1.54  24.37
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## gdp_growth.l1  -0.0136     0.0609   -0.22   0.8240
## inflation.l1   -0.2689     0.0839   -3.21   0.0015 **
## d_FEDFUNDS.l1   1.3542     0.2971    4.56  8.1e-06 ***
## const           3.9914     0.4340    9.20  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 '
##
## Residual standard error: 4.04 on 250 degrees of freedom
```

Diagnostic Check

- Conduct a Portmanteau test for serial correlation in the residuals.
- Keep adding lags until the test fails to reject the null of no serial correlation.

```r
serial.test(var.fit, lags.pt = 12, type = "PT.asymptotic")
```

```
##
##  Portmanteau Test (asymptotic)
##
## data:  Residuals of VAR object var.fit
## Chi-squared = 166, df = 99, p-value = 3e-05
```

Add one more lag and re-test:

```r
var.fit2 <- VAR(macro_st, p = 2, type = "const")
serial.test(var.fit2, lags.pt = 12,
type = "PT.asymptotic")
```

```
##
##  Portmanteau Test (asymptotic)
##
## data:  Residuals of VAR object var.fit2
## Chi-squared = 128, df = 90, p-value = 0.005
```

Step 4: Lag Selection — Decision

Summary

Model	χ^2	p**-value**	**Decision**
VAR(1)	166	0.000	Reject H_0 — serial correlation present
VAR(2)	128	0.005	Reject H_0 — serial correlation present
VAR(3)	95	0.100	Fail to reject H_0 — residuals are white noise

- VAR(1) and VAR(2) leave predictable structure in the residuals.
- VAR(3) resolves the problem — consistent with the AIC and FPE recommendation.

We proceed with $p = 3$. BIC's preference for parsimony was overruled by the data — a common outcome in macro applications.

Estimated Model

```r
var.fit3 <- VAR(macro_st, p = 3, type = "const")
summary(var.fit3)
```

```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: gdp_growth, inflation, d_FEDFUNDS
## Deterministic variables: const
## Sample size: 252
## Log Likelihood: -1498.763
## Roots of the characteristic polynomial:
## 0.877 0.578 0.578 0.545 0.545 0.483 0.483 0.325 0.325
## Call:
## VAR(y = macro_st, p = 3, type = "const")
##
##
## Estimation results for equation gdp_growth:
## ===========================================
## gdp_growth = gdp_growth.l1 + inflation.l1 + d_FEDFUNDS.l1 +
```

Reading the VAR Output

Each equation is estimated by OLS. The summary provides:

- **Coefficient estimates** — entries of each $\hat{\mathbf{A}}_\ell$
- **Standard errors and t-statistics**
- R^2 — fit for each equation separately
- **Residual standard error**

Two things to keep in mind:

1. Individual coefficient significance is of limited interest in a VAR. The system matters more than any single coefficient — use Granger causality tests for inference on predictive content.
2. Parameters grow fast: a VAR(p) with K variables has $K^2p + K$ parameters.

- With $K = 3$, $p = 2$: **21 parameters**.
- With $K = 3$, $p = 3$: **30 parameters**.

Step 5: Diagnostic Checking (Again)

```
## Serial correlation -- Portmanteau test
serial.test(var.model.fit, lags.pt = x,
            type = "PT.asymptotic")

## Normality of residuals
normality.test(var.model.fit)

## Stability -- roots of companion matrix (all should be < 1)
vars::roots(var.model.fit)
```

Stability Check

All roots of the companion matrix should lie **inside** the unit circle.

```
vars::roots(var.fit3)
```

```
## [1] 0.8774 0.5779 0.5779 0.5449 0.5449 0.4828 0.4828 0.3248
```

- All roots are strictly less than 1.

The stability condition

- If **all** roots < 1, the VAR is **stable** — shocks decay over time and the system returns to its mean.
- A root at or above 1 would indicate an explosive or integrated process, making IRFs and forecasts unreliable.
- The largest root is 0.877 — close to 1, indicating **high persistence**. Shocks to this system decay slowly.

Step 5: Diagnostic Checking

What we are checking:

- **Serial correlation** — VAR residuals should be white noise. A significant Portmanteau test suggests the lag order is too short.
- **Normality** — Required for valid small-sample IRF inference. Violations are common with macro data; bootstrap confidence intervals are a robust alternative.
- **Stability** — All roots of the companion matrix should lie inside the unit circle. If not, the VAR is explosive and results are unreliable.

If serial correlation is detected

Increase p and re-estimate. Re-run `VARselect()` with a higher `lag.max` if needed.

Section 6

Granger Causality

Granger Causality

Definition

y_2 **Granger-causes** y_1 if past values of y_2 help predict y_1, *over and above* the information already contained in past values of y_1 itself.

In the VAR, y_2 does *not* Granger-cause y_1 if:

$$a_{12,1} = a_{12,2} = \cdots = a_{12,p} = 0$$

i.e., all coefficients on lags of y_2 in the y_1 equation are jointly zero.

This is a Wald (think joint F test) test on a block of coefficients.

$$H_0 : a_{12,1} = \cdots = a_{12,p} = 0 \quad (y_2 \text{ does not Granger-cause } y_1)$$

Reject H_0 if p-value $< \alpha$ — conclude y_2 **does** Granger-cause y_1.

Granger Causality

Important caveats:

- Granger causality is about **predictive content**, not structural causation. A variable can Granger-cause another purely because it is a good leading indicator.
- The result is conditional on what else is in the system.
- Reverse causality is possible and common.

Economic interpretation

Does the fed funds rate Granger-cause output growth? Does inflation Granger-cause the fed funds rate (a Taylor rule implication)? These are empirically testable propositions with our system.

Granger Causality in R

```r
## Does inflation Granger-cause GDP growth and d_FEDFUNDS?
vars::causality(var.fit3, cause = "inflation")$Granger

## Does d_FEDFUNDS Granger-cause GDP growth and inflation?
vars::causality(var.fit3, cause = "d_FEDFUNDS")$Granger
```

```
## --- Does inflation Granger-cause the other variables? ---

##
##  Granger causality H0: inflation do not Granger-cause gdp_g
##  d_FEDFUNDS
##
## data:  VAR object var.fit3
## F-Test = 2.3, df1 = 6, df2 = 726, p-value = 0.03

##
## --- Does d_FEDFUNDS Granger-cause the other variables? ---

##
##  Granger causality H0: d_FEDFUNDS do not Granger-cause gdp_
##  inflation
##
## data:  VAR object var.fit3
## F-Test = 7.4, df1 = 6, df2 = 726, p-value = 1e-07
```

Note: `causality()` tests whether the `cause` variable has no predictive content for *all* remaining variables jointly. Pairwise tests require individual equation F tests.

Granger Causality — Results

Summary

Hypothesis	**F-stat**	p**-value**	**Decision**
Inflation $\not\rightarrow$ GDP growth, ΔFFR	2.3	0.03	Reject H_0
ΔFFR $\not\rightarrow$ GDP growth, inflation	7.4	< 0.001	Reject H_0

- Past inflation has predictive content for the rest of the system.
- Past changes in the fed funds rate have strong predictive content — the largest F-statistic of the two.

Economic interpretation

Both results are intuitive. Inflation signals future monetary tightening (a Taylor rule implication), while changes in the fed funds rate transmit to output and inflation with a lag — the classic monetary policy transmission mechanism.

- We will see *how* these effects propagate, and over what horizon, when we turn to IRFs in Part 2.

Section 7

Forecasting

Step 6: Forecasting with the VAR

The h-step-ahead forecast is constructed recursively:

$$\hat{\mathbf{y}}_{T+h|T} = \hat{\mathbf{c}} + \hat{\mathbf{A}}_1 \hat{\mathbf{y}}_{T+h-1|T} + \cdots + \hat{\mathbf{A}}_p \hat{\mathbf{y}}_{T+h-p|T}$$

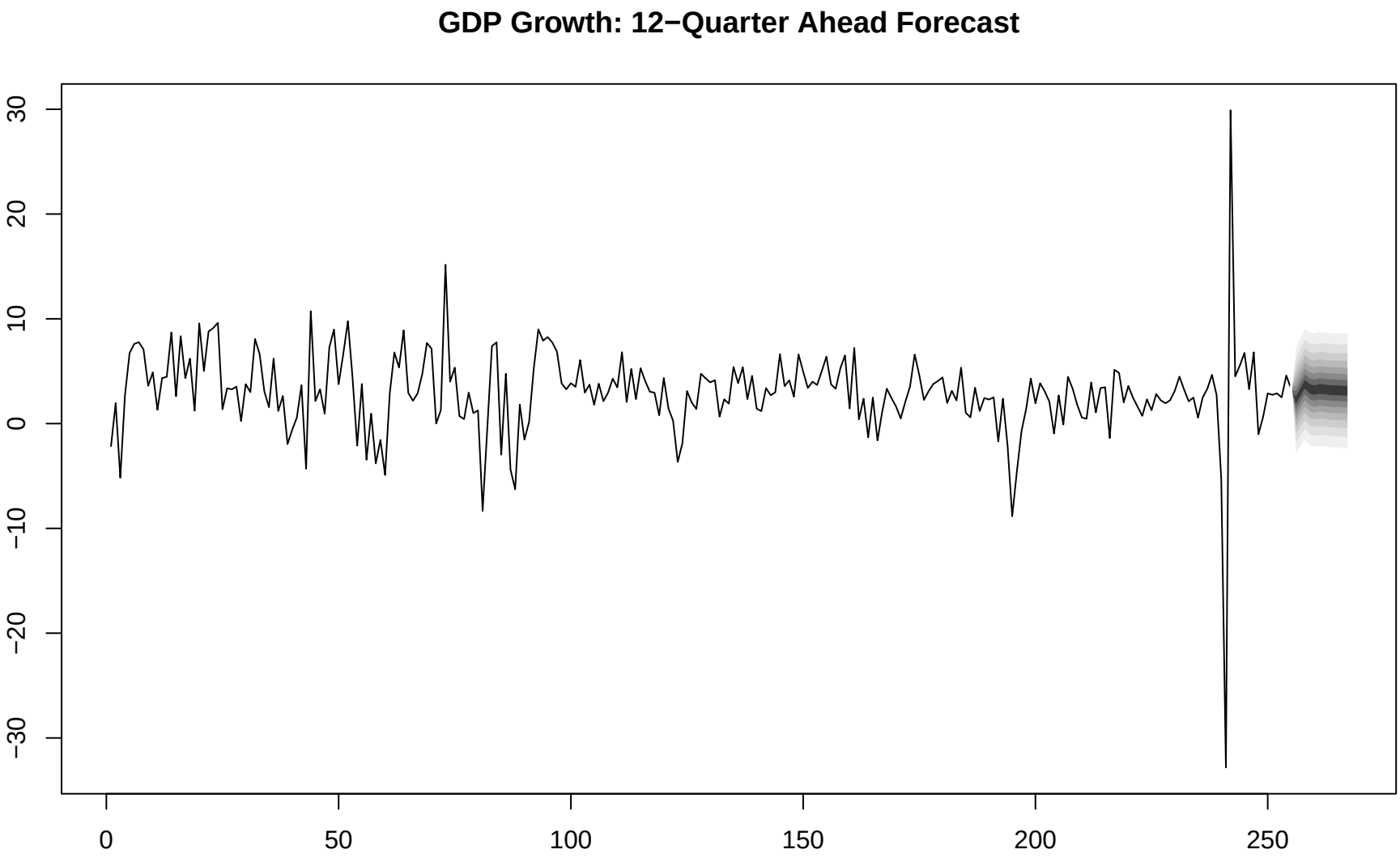
where $\hat{\mathbf{y}}_{T+j|T} = \mathbf{y}_{T+j}$ for $j \leq 0$ (observed values used directly).

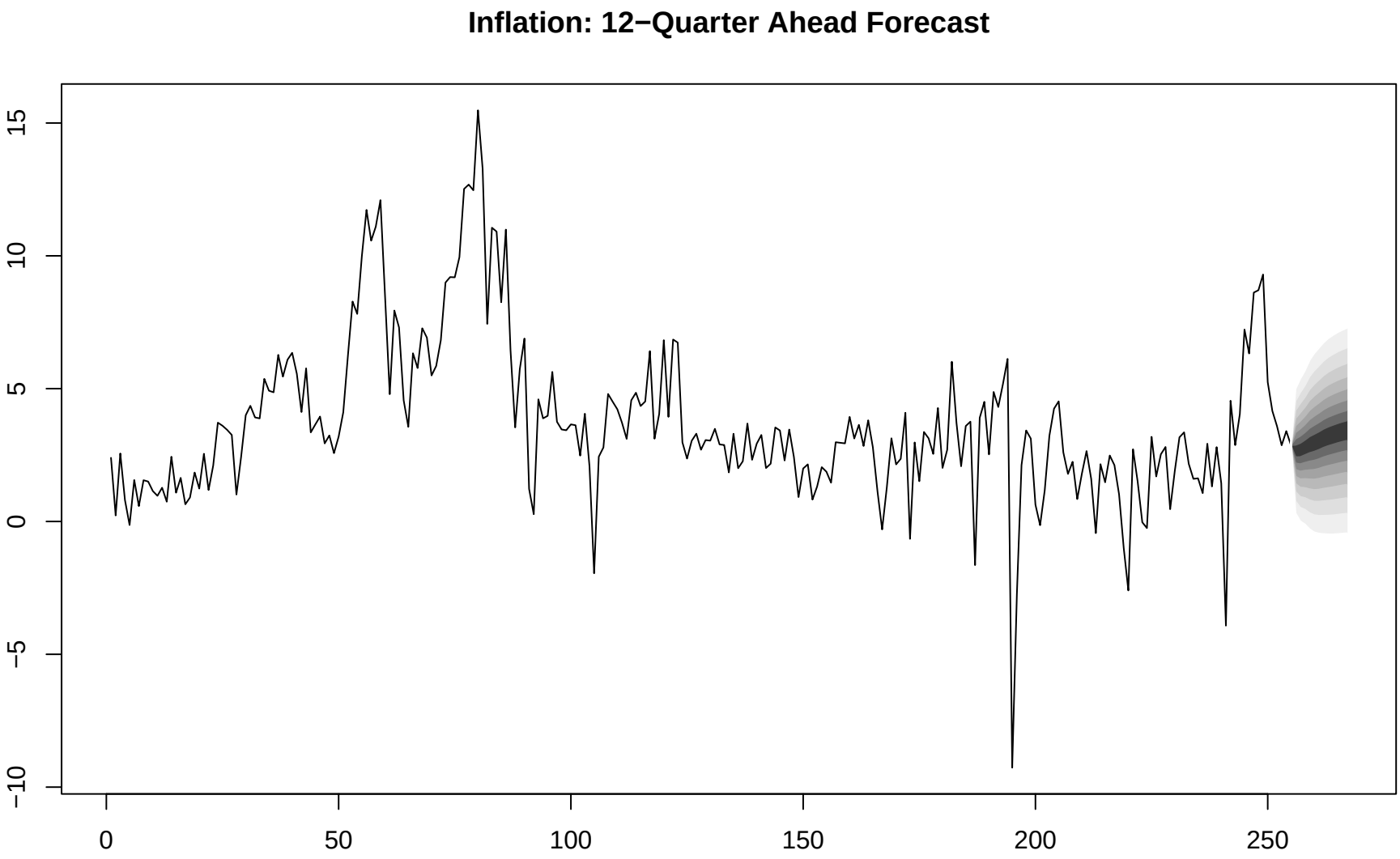
Forecast uncertainty

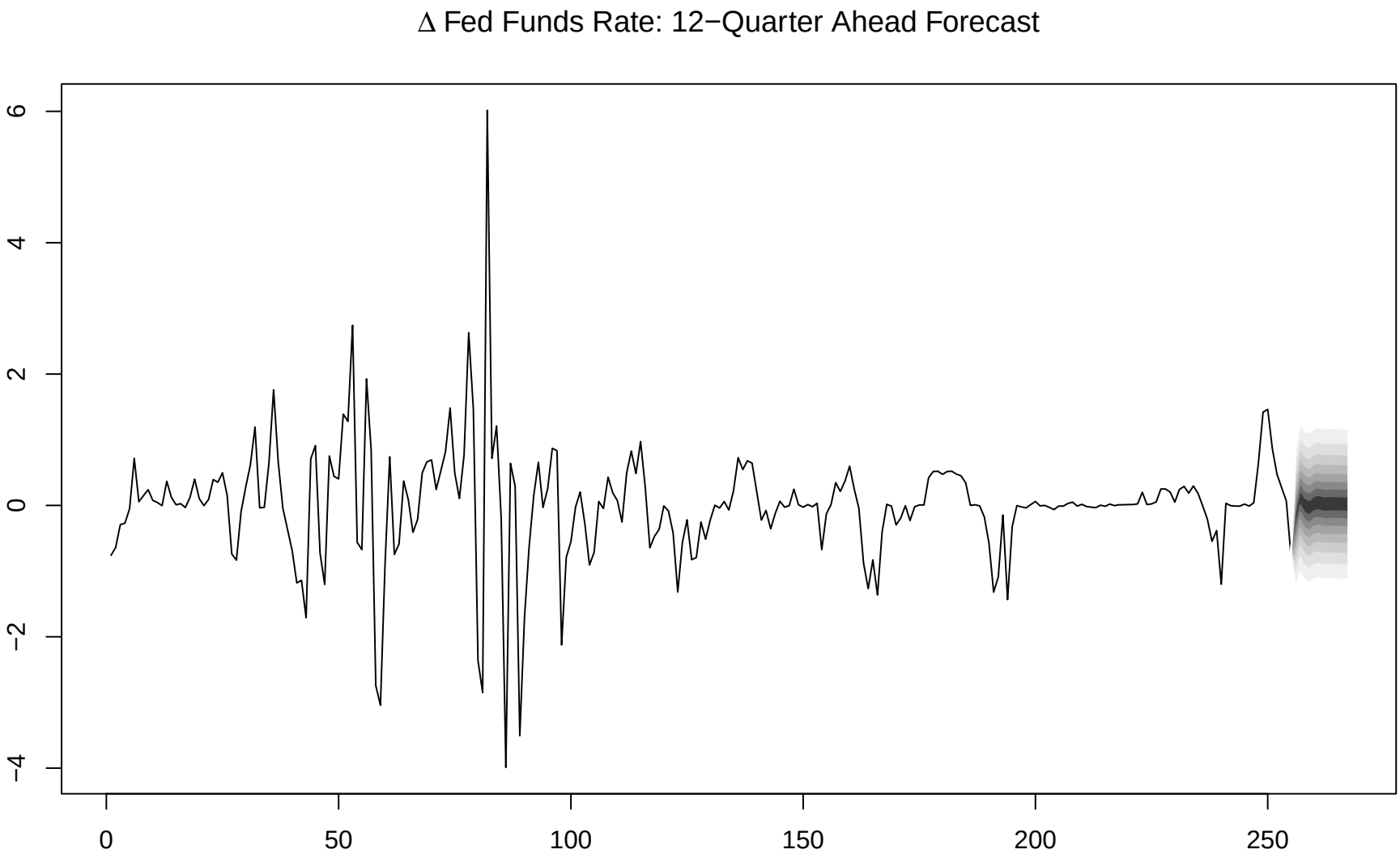
Forecast intervals widen with the horizon and account for innovation variance ($\boldsymbol{\Sigma}$). Parameter estimation uncertainty is typically ignored in practice.

Forecasting with the VAR

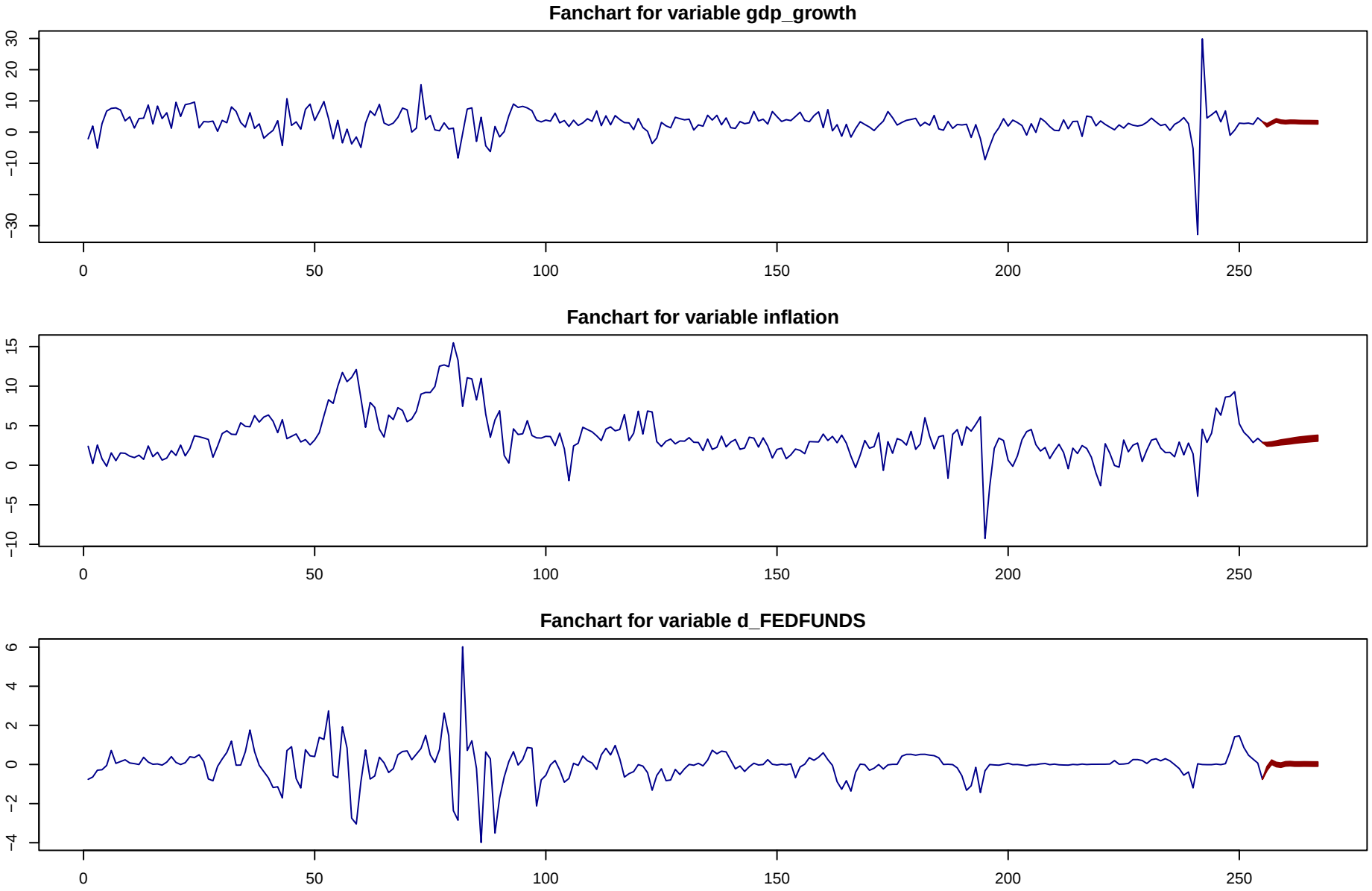
```r
var.fc <- predict(var.fit3, n.ahead = 12,
                          ci = 0.95)

vars::fanchart(var.fc, names = "gdp_growth",
         main = "GDP Growth: 12-Quarter Ahead Forecast")
```


GDP Growth: 12−Quarter Ahead Forecast
30
20
10
0
−10
−20
−30
0
50
100
150
200
250


Inflation: 12−Quarter Ahead Forecast
15
10
5
0
−5
−10
0
50
100
150
200
250


Δ Fed Funds Rate: 12−Quarter Ahead Forecast
6
4
2
0
−2
−4
0
50
100
150
200
250

```r
vars::fanchart(var.fc, plot.type = "multiple",
               col.y = "blue4", col= "darkred",
               mar = c(3, 3, 2, 1))
```



Looking Ahead

Part 1 Summary

1. **Motivation** — why systems models are needed
2. **Specification** — the VAR(p) model
3. **Stationarity** — unit root testing and the levels vs. differences question
4. **Lag selection** — AIC, BIC, HQ
5. **Estimation** — OLS equation by equation
6. **Diagnostics** — serial correlation, normality, stability
7. **Granger causality** — predictive content testing
8. **Forecasting** — recursive multi-step forecasts and fan charts

Part 2: Impulse Response Functions, Forecast Error Variance Decomposition, and the identification problem that motivates Structural VARs (SVAR).